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Zeolite shape selectivity impact on LDPE and PP catalytic pyrolysis products and coke nature†

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Catalytic pyrolysis of plastic waste can offer an economical process for plastic waste conversion due to the capability to produce high-quality products with high yields. A simple stirred tank reactor was used for two significant types of polyolefin (low-density polyethylene and polypropylene) under thermal and catalytic conditions, *i.e.*, promoted by HZSM-5, HBEA, and HY. Complete conversion of the polymers into light gases, aromatic coke, and liquid products (mainly in the gasoline range) was achieved in the presence of all catalysts. The detailed analysis of the gas, liquid, and coke deposited inside the zeolite framework was used to unravel the mechanism responsible for converting LDPE and PP over the different zeolites. LDPE and PP particular polymer structures significantly impacted the reaction pathway, with PP yielding more reactive and stereochemically cumbersome intermediates. Furthermore, secondary reactions, particularly hydrogen transfer, secondary cracking, cyclization, and oligomerization, significantly impacted the product distribution. The higher acidity and particular framework shape of HZSM-5 benefited secondary cracking and hydrogen transfer reaction leading to higher selectivity towards gas and aromatic compounds. Further, HBEA and HY, respectively, favored isomerization and cyclization instead. Additionally, the spent catalysts' physicochemical properties combined with the information obtained from the analysis of coke composition were used to differentiate the processes responsible for the deactivation of the different zeolites. The presence of polyaromatic coke promoted the deactivation of the catalysts, with HZSM-5 retaining more than 50% of its acidity after the reaction. In contrast, more than 80% of HY and HBEA acid sites were lost under the same conditions.

1. Introduction

Worldwide, there is increasing public and political pressure to develop circular economy strategies that will deliver improved sustainability for products and processes; according to public opinion, the generation and inappropriate disposal of plastic waste is currently one of the biggest environmental threats.[1] Despite this, plastic production is likely to increase steadily, if not exponentially, over the coming decades if no changes are made to our current patterns of consumption.[2,3] The development of new waste management alternatives capable of diverting plastic waste, particularly mixed and contaminated waste, from landfills and from leakage to the environment, *e.g.*, oceans, would benefit both the environment and the economy. From an environmental point of view, the pyrolysis of plastics is an interesting and mature technology capable of producing high-quality fuels or olefins.[4] Even though not mandatory, the use of a catalyst significantly improves the quality and yield of fuel[5,6] or olefins.[7–9] Additionally, catalytic pyrolysis can reduce the reaction temperature and time, particularly when compared to thermal cracking.[10] Such improvements can make the process economic through lower operating costs and better-quality products.

Multiple catalysts have been used to promote the conversion of plastics into fuels and olefins, *e.g.*, FCC e-cat,[11,12] MCM-41,[10,13,14] ZSM-5,[8,15–19] BEA,[16,20] zeolite Y.[16,21,22] The acid properties of zeolites make these materials, and their long-term use in converting heavy hydrocarbons, *i.e.*, crude oil into fuels, the catalyst of choice for converting plastic waste, especially polyolefin waste. Moreover, the hydrocarbon structure of polyolefin polymers and their derived pyrolysis products is similar to that of other refinery hydrocarbons, *i.e.*, linear and branched paraffins, olefins, aromatics, *etc.*, which have been extensively studied in the literature for many years. Therefore, the catalytic

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pyrolysis of polyolefins over zeolite would be expected to proceed *via* similar reaction mechanisms as those of smaller hydrocarbons, *i.e.*, catalytic cracking, hydrogen transfer, isomerization condensation, *etc.* Indeed, the similarity between polyethylene and polypropylene and the typical feedstocks of FCC units has led to multiple studies assessing co-feeding in refineries.[23–26]

The choice of the reactor also plays an important part in the pyrolysis, with the majority of the reported work being done in fixed, fluidized, or spouted bed reactors.[27] Yet, a significant number of commercial plants focused on converting polymer waste into fuels and olefins are based on a stirred tank or rotary kiln technologies.[28,29] Such commercially practiced technologies can easily handle molten plastic feedstocks. However, while these processes have been well-studied for thermal pyrolysis, they are yet to be adapted for catalytic pyrolysis. Serrano *et al.* have studied a screw-kiln reactor for the catalytic pyrolysis of polyethylene, observing an alteration of the product spectrum caused by the longer residence time.[14] Indeed, the stirred tank and kiln reactors can have significantly longer residence times (5–30 s) than fluidized bed reactors (<2 s),[30,31] favoring the occurrence of secondary reactions, *e.g.*, hydrogen transfer, condensation, secondary cracking, *etc.* The study of the catalytic conversion of waste plastics in conditions similar to those practiced in commercial-scale thermal units, which have higher product residence time, would help in a faster transition into more efficient technologies capable of producing higher-value products from plastic waste.

In this work, the conversion of low-density polyethylene (LDPE) and polypropylene (PP) in the presence of HY, HBEA, and HZSM-5 zeolites will be performed in a stirred tank reactor. First, the yield of gas and liquid products from the reaction were quantified and their composition was characterized by chromatography and qualitative 2D NMR (HSQC – for the liquid product). Additionally, the catalyst deactivation was studied *via* assessment of the degree of modification in the spent zeolites' textural properties, using nitrogen sorption. Further, the heavy molecules present in the coke inside the zeolite pores were extracted after mineralization in hydrofluoric acid (HF), identified by GC/TOF-MS, and quantified. Finally, the overall distribution of the products from the reaction, *i.e.*, gas, liquid, and coke, was used to characterize the conversion mechanisms of the two polymers and respective pyrolysis products according to each zeolite framework.

2. Experimental

2.1. Materials

Unpigmented low-density polyethylene (LDPE) commercially known as Alkathene LDH210 from Qenos, Australia, was used in this study. LDPE came in the form of beads with a 3–4 mm diameter and density range of 0.91–0.945 g cm^{-3}. Polypropylene (PP) feed material was also a commercial-grade product called "Moplen HF 500N" from Lyondellbasell. PP came in 1 mm particles with a density of 0.90 g cm^{-3}. The amount of feed material taken for each experiment was 30 g.

The zeolites ZSM-5 (CBV 2314), BEA (CP814E*), and Y (CBV 720) from Zeolyst International were selected for this study. ZSM-5 and BEA zeolites were purchased in the ammonium-modified form, *i.e.*, NH_4ZSM-5 and NH_4BEA. Therefore, they were first calcined at 550 °C for 4 h to transform them into acid form, *i.e.*, HZSM-5 and HBEA, before use. All of the catalysts were used as powder. A 10 wt% loading of catalysts (*i.e.* 3 g) was taken in each experiment.

2.2. Physicochemical characterization of catalyst

A Micromeritics ASAP 2020 apparatus was used to perform N_2 sorption isotherms at −196 °C to determine the textural properties of catalysts. All catalyst samples were degassed for 6 hours at 200 °C under vacuum. The Brunauer–Emmett–Teller (BET) method was used to estimate the surface area. The total pore volume (V_{total}) was determined by using the volume of nitrogen adsorbed at P/P_0 of 0.97, the micropore volume (V_{micro}) was obtained by using the t-plot method, and the mesopore volume (V_{meso}) corresponds to the difference between V_{total} and V_{micro}.

The quantification of Brønsted and Lewis sites in parent and spent zeolite samples was performed by pyridine adsorption followed by infrared spectroscopy. Before pyridine adsorption, the parent samples were pretreated at 400 °C under air flow of 60 mL min^{-1} for 2 h and degassed at 10^{-5} bar for 1 h at 150 °C. On the other hand, the spent zeolites were solely outgassed at 10^{-5} bar at 150 °C for 2 h to avoid the desorption of organic compounds. At 150 °C, the samples were exposed to 2 mbar of pyridine for 5 min, after which the sample was outgassed for 1 h, at 10^{-5} bar, at 150 °C, to remove all physisorbed pyridine. The Beer–Lambert–Bouguer law was applied according to the formula ($[\text{PyH}^+]$ and $[\text{PyL}] = \left(\frac{S}{m}\right)\frac{A}{\varepsilon}$, where $[\text{PyH}^+]$ and $[\text{PyL}]$ are the concentrations of Brønsted and Lewis acid sites in (μmol g^{-1}), respectively; A is the area of the characteristic adsorption band for pyridine adsorbed on Brønsted (1545 cm^{-1}) and Lewis (1455 cm^{-1}); m is the wafer mass; S is the wafer surface, *i.e.*, 2 cm^2; and, ε is the integrated molar adsorption coefficient: $\varepsilon_{1545} = 1.13$ and $\varepsilon_{1455} = 1.28$ cm mol^{-1}).[32] All FTIR spectra were obtained at room temperature. A similar procedure was used for the quantification of Brønsted and Lewis sites in the spent zeolite samples, with the spent samples being pretreated at 150 °C, at 10^{-5} bar for 1 h.

2.3. Plastic pyrolysis

Pyrolysis experiments were carried out under semi-batch conditions using a quartz stirred tank reactor (500 mL), schematically represented in Fig. 1. The feed materials along with the catalyst were introduced at room temperature and heated up to a temperature of 450 °C over 7 min and held for 30 min, under a continuous flow of N_2 at 150 mL min^{-1}. The reaction mixture was continuously stirred to enhance contact between plastic and catalyst and to favour the desorption of reaction products. The liquid and gas products from this pyrolysis passed through a glass condenser, which was placed in an ice bath, *i.e.*, ∼0 °C. The liquid was collected after reaction and analyzed *ex situ*. It should be noted that no solvent was used to

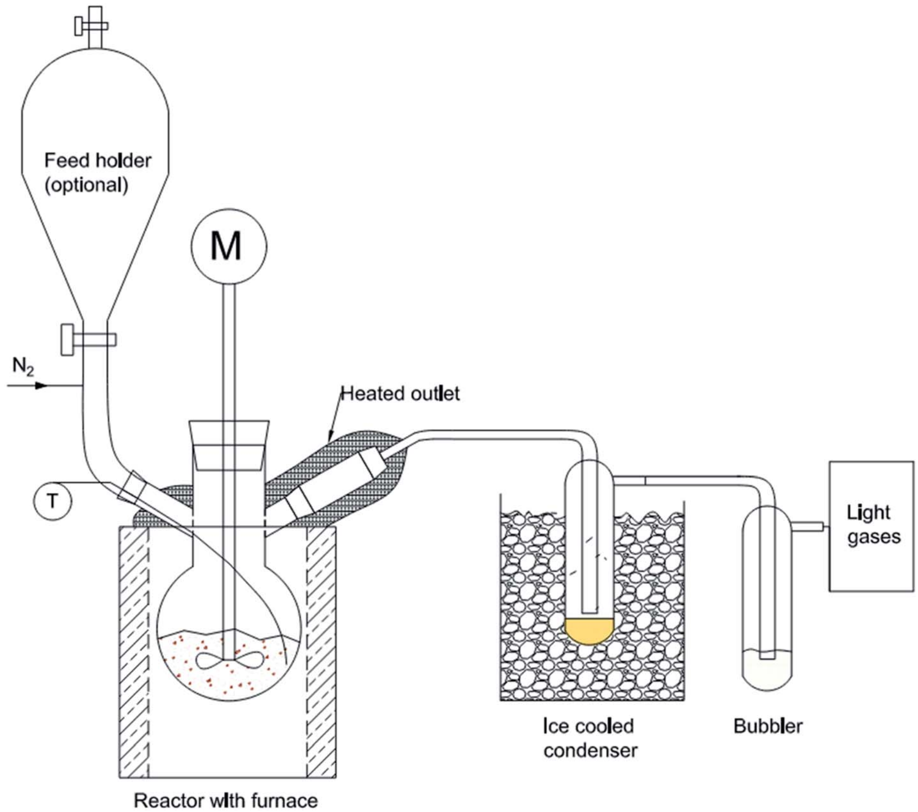


Fig. 1 Experimental setup (representative stirred tank reactor) used to study the cracking behaviour of polymers over zeolites. M = motor for stirrer T = thermocouple.

capture the condensation product. After the condenser, the non-condensed products passed through a bubbler to capture any remaining condensable products, and the non-condensable gases were collected in gas bags for analysis. It should also be noted that the amount of trapped liquid product captured in the bubbler was less than 3 wt% of feed. The sample collected from the ice-cooled condenser (without any use of solvent) was designated as the liquid product for the pyrolysis. The reactor, along with all its parts, was weighed before and after the experiment to determine the amounts of various products.

2.4. Pyrolysis product analysis

A Shimadzu gas chromatograph (GC-2014) equipped with a thermal conductivity detector (TCD) was used to analyze the gas samples. The GC was fitted with a ShinCarbon Packed column (Restek) with the specification of 2 m length and 2 mm internal diameter. The gas product identification and quantification was made using a gas standard mixture with hydrocarbons ranging from C_1 to C_3.

The carbon distribution of the liquid samples was determined using a Shimadzu GC-2010 gas chromatograph that was fitted with an HP-ULTRA-1 column with dimensions of 25 m length, 0.20 mm internal diameter, and 0.33 μm film thickness. A flame ionization detector (FID) was used for the detection and quantification of the products. Samples from the catalytic pyrolysis of polyolefin were injected directly, without any dilution, as they were in liquid form at room temperature. However, the samples from two thermal pyrolysis experiments, which were solid at room temperature, were diluted in cyclohexane before injection in the GC-FID. The peak for solvent was excluded from the analysis.

Qualitative 2D ^{1}H–^{13}C HSQC (Heteronuclear Single Quantum Correlation) NMR analysis of the liquid products was performed on a Bruker Avance 400 MHz. For each analysis, 125 mg of liquid sample was dissolved in 300 μL of deuterated chloroform ($CDCl_3$). Independent zones (Fig. S1 and Table S1†) in the NMR spectra were used to identify the relative proportions of aliphatic CH, CH_2, and CH_3, allylic C=CCH, alkyl $C_{aromatic}$–CH, vinylic C=CH, and aromatic $C_{aromatic}$–H bonds.

2.5. Coke characterization

The amount of coke deposited on the catalyst was determined by thermogravimetry on an SDT Q600 TA apparatus. The spent catalysts were first heated to 50 °C for 10 min and then to 800 °C for under an air flow (100 mL min^{-1}), at a heating rate of 10 °C min^{-1}. During this oxidative treatment, the sample mass was recorded, with the coke amount in the spent catalyst corresponding to the weight loss between 300 °C and 800 °C.

The investigation of the chemical composition of the coke found inside the zeolite porous structure was derived from an established methodology developed by Magnoux *et al.*[33] First, the soluble coke in the zeolite's external surface, identified as external coke, was extracted by washing the catalyst with dichloromethane (CH_2Cl_2) at 110 °C under 10 MPa N_2 using a Dionex ASE 350 apparatus (ThermoFisher, Illkirch). Next, the coke molecules trapped inside the zeolite pores were extracted through the mineralization of the catalyst, with a 51% hydrofluoric acid (HF) solution at room temperature, followed by CH_2Cl_2 to the solution for liquid–liquid extraction. After the aluminosilicate matrix dissolution and liquid–liquid extraction, the acid solution was neutralized with sodium hydrogen carbonate ($NaHCO_3$). Any insoluble coke, which accounted for less than 5 wt% of the total coke, was recovered by filtration. The coke fraction soluble in CH_2Cl_2 after the zeolite mineralization is denominated as soluble coke.

The coke molecules from the zeolite pores soluble in CH_2Cl_2 after zeolite mineralization (soluble coke) and the external cokes were characterized by GC Agilent 7890 A equipped with a TOF MS Thermo Electron DSQ. Additionally, the soluble coke composition was quantified by GC-FID using a similarly configured apparatus. The analysis of insoluble coke products was not performed due to the low contribution of this fraction to the overall coke composition.

3. Results and discussion

3.1. Catalyst characterization

The physicochemical and textural properties of the commercial catalysts are shown in Table 1. The textural properties of the catalysts, determined by nitrogen physisorption, reflected the differences in the zeolite microporous frameworks. HZSM-5 has the smallest micropores among the three tested zeolites (Table 1), reflecting on this sample having the smallest micropore volume (0.150 $cm^3\ g^{-1}$). By contrast, HY zeolite has the largest micropore volume (0.236 $g\ cm^{-3}$) and framework pore size of the three catalysts used. Unlike micropore volume, mesopore volume is not linked to the zeolites framework. Instead, the presence of mesopores in zeolite catalysts can be explained by small spaces between crystals, particularly in nano-crystal zeolites, or by defects created on the zeolites crystallite.[34] When comparing the crystallite size of the tested zeolites, the crystal diameter of HBEA is 20–30 nm,[35] whereas HZSM-5 and HY zeolite crystal sizes are significantly bigger, *i.e.*, >200 nm. Consequently, the significantly higher mesopore volume of HBEA (0.730 $cm^3\ g^{-1}$) can be explained by the mesopores existing in between the small crystallites.

Table 1 Physicochemical and textural properties of catalysts

	HZSM-5	HBEA	HY
Si/Al	12	13	15
Surface$_{BET}$ ($m^2\ g^{-1}$)	506	710	733
$V_{total}{}^a$ ($cm^3\ g^{-1}$)	0.257	0.908	0.496
$V_{meso}{}^b$ ($cm^3\ g^{-1}$)	0.107	0.730	0.260
V_{micro} ($cm^3\ g^{-1}$)	0.150	0.178	0.236
[PyH^+] (μmol g^{-1})	505	477	272
[PyL] (μmol g^{-1})	149	463	101
Framework pore size (Å)[41]	5.1 × 5.5; 5.3 × 5.6	6.6 × 6.7; 5.6 × 5.6	7.4 × 7.4
Max molecular inclusion^c (nm)	0.636	0.668	1.124

^{a} Total volume calculated for $P/P_0 = 0.99$. ^{b} $V_{meso} = V_{total} - V_{micro}$. ^{c} Largest sphere size which can be included inside the framework.[41]

The Si/Al ratio of these catalysts was chosen to be as close to each other as possible for comparison purposes and ranged from 12 for HZSM-5 to 15 for HY zeolite. Despite the small range of Si/Al, the zeolites displayed significant differences in terms of acidity (Table 1). While HZSM-5 and HBEA Brønsted acidity ([PyH^+]) was equivalent, *i.e.*, 505 μmol g^{-1} *vs.* 477 μmol g^{-1}, respectively, HY contained significantly fewer Brønsted sites (272 μmol g^{-1}). Further, the Lewis acid site density ([PyL]) of HY (101 μmol g^{-1}) and HZSM-5 (101 μmol g^{-1}) was considerably smaller than that of HBEA (463 μmol g^{-1}). In general, the total amount of acid sites (Lewis + Brønsted) in a zeolite is proportional to the amount of aluminum in the sample, thus explaining the lower acidity of HY. Yet, HBEA contains significantly more Lewis sites than HZSM-5 while displaying similar Brønsted acidity and slightly lower Si/Al. The presence of sodium ions in HZSM-5 could explain such a discrepancy, as these are well known to neutralize Brønsted acid sites.

In zeolites, Brønsted acid sites are typically associated with framework coordinated aluminum, while Lewis acid sites are related to extra-framework aluminum species (EFAl).[36] Therefore, the higher Lewis acidity observed in HBEA can be explained by a higher concentration of EFAl species in this sample. It is important to mention that Brønsted acid sites can promote catalytic cracking of hydrocarbons, like LDPE and PP, more easily than Lewis sites.[37,38] Yet, several studies have demonstrated that the presence of EFAl in proximity to Brønsted sites can significantly enhance these active sites' activity.[39,40]

3.2. Polyolefin pyrolysis

3.2.1. Product distribution. The three commercial zeolites were used to promote the cracking of polyethylene and polypropylene during pyrolysis, with the presence of the catalyst significantly improving the conversion of the polymer resins (Fig. 2). At 450 °C, the thermal conversion of LDPE and PP in the absence of catalyst yielded 36.7 wt% and 34.9 wt% of residue, respectively. This residue corresponds to a high molecular weight product, which, at the temperature of 450 °C, stays in the reactor and is not converted into gas or lighter liquid. The use of zeolite catalysts, by contrast, resulted in no visible residue. Yet, the solid material deposited on the zeolites, represented 1.8 wt%, 1.3 wt%, and 0.4 wt% for HY, HBEA and HZSM-5 respectively, independent of the feedstock. It is important to mention that the solid product obtained during catalytic cracking of LDPE and PP was mainly caused by coke formation in the catalyst and not by the unconverted polymer, as will be discussed later. The higher coke yield on HY was directly related to the larger pore size of this zeolite, which favours hydrogen transfer reactions and, consequently, coke formation.[33,42,43] By contrast, the smaller pores of ZSM-5 zeolite hinder the formation of bulkier coke molecules.[33,43] Hence, reducing the formation of coke in this zeolite.

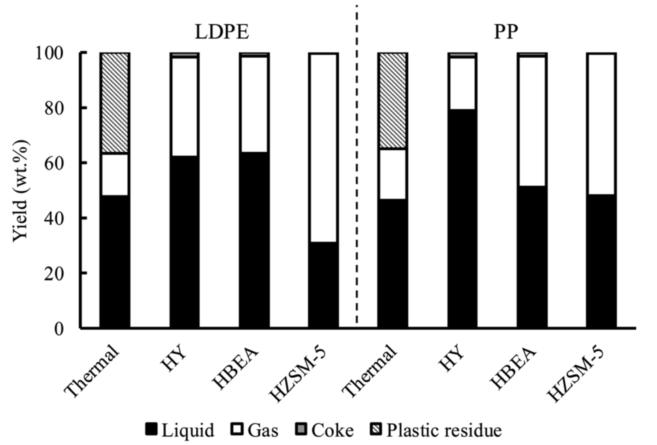


Fig. 2 Yields for thermal and catalytic pyrolysis of LDPE and PP over HY, HBEA, and HZSM-5 zeolites.

Under thermal cracking conditions, product yields from LDPE and PP were very similar, with the yields of liquid and gas products from LDPE and PP being 47.8 wt% and 46.3 wt%, and 15.6 wt% and 18.9 wt%, respectively. The similarity between the thermal cracking behaviour of the two polymers could be explained by the similar C–C bond dissociation energy of LDPE and PP, *i.e.*, 341.1 kJ mol^{-1} and 334.1 kJ mol^{-1}, respectively.[15] Additionally, Sørum *et al.* studied the kinetics for the thermal pyrolysis of waste plastics, which showed that the activation energies for LDPE and PP were 340.8 kJ mol^{-1} and 336.7 kJ mol^{-1}, respectively.[44]

The acid catalyst properties of the zeolites enhanced the cleavage of the C–C bonds of LDPE and PP, leading to an increase in liquid and gas products. Despite all three zeolites displaying acid properties, the product selectivity was significantly affected by each catalyst's properties (Table 1), with the selectivity also differing between LDPE and PP as well (Fig. 2). By comparing gas yields, HY and HZSM-5 showed higher selectivity towards gas from LDPE, but HBEA produced more gas from PP. On the other hand, HBEA led to similar gas and liquid yields to HY for the conversion of LDPE, but the performance of this zeolite was similar to HZSM-5 when converting PP. The difference in oil and gas yields when using the same catalyst for different polymers, along with the different cracking behaviours observed with each zeolite catalyst, suggests that the properties of the polymer and the interaction behaviour of the polymer and intermediates with zeolite catalysts also affects the cracking behaviour.

HZSM-5 zeolite was particularly selective for the formation of gas products (Fig. 2). When used for the cracking of LDPE, HZSM-5 yielded 69.1 wt% of gaseous products and 30.9 wt% of liquid. Alternatively, for PP, HZSM-5 generated 51.9 wt% of gas and 48.1 wt% of liquid. Independent of the polymer feedstock, the coke formation was lower than 0.1 wt%. The high cracking selectivity of HZSM-5 zeolite for gas-phase products in the conversion of plastics and crude oil is often referred to in literature.[11–14] The selectivity of HZSM-5 zeolite to light products is attributed to a combination of high acidity (Table 1) and low coking rates, which is a consequence of shape selectivity provided by the MFI's 3D medium-sized pore framework, which delays catalyst deactivation.[16]

While HZSM-5 was always particularly selective to gaseous products, the yield reduced from 69.1 wt% to 51.9 wt% when PP replaced LDPE. The lower selectivity towards light products during PP cracking suggests a limitation in the capacity of HZSM-5 for C–C bond scission, even though acid-promoted catalytic cracking is favoured in branched *versus* linear hydrocarbons.[45,46] Zhou *et al.* attributed the lower potential of HZSM-5 to promote cracking in PP *versus* LDPE to the limited diffusion of the reaction intermediates inside the zeolite framework,[15] *i.e.*, shape selectivity. According to the authors, the linear LDPE and its decomposition products can diffuse quickly inside HZSM-5. By contrast, the diffusion of the isomerized structure of PP and its degradation products is limited. Indeed, the restricted diffusion and shape selectivity of isomerized hydrocarbons inside the HZSM-5 framework has been reported before.[47–49] Yet, enhanced coke formation results from diffusion limitations, which favour condensation reactions. Therefore, the low and near-identical coke yield observed over HZSM-5 for LDPE and PP conversion indicates limitations in the intermediates products diffusion inside the HZSM-5 framework should not be responsible for the observed results, as discussed in the following sections.

For cracking over HY, 78.8 wt% liquid was produced from PP, whereas 62.1 wt% was produced from LDPE. HY displayed the lowest acidity (Table 1), which could explain the high liquid yield, *i.e.*, lower cracking rate. Still, the higher selectivity towards liquid products for PP *versus* LDPE indicates acid site concentration might not be controlling the product, as branched hydrocarbons undergo cracking more easily.[45,46] Therefore, another phenomenon is interfering with the polymers cracking behaviour over HY. However, unlike HZSM-5, the framework structure of HY allows for the diffusion of much larger molecules (Table 1).[50] An alternative explanation for the higher liquid yield obtained for catalytic pyrolysis of PP could be the oligomerization of smaller intermediate products, *i.e.*, small olefins combining into liquid range products. This secondary bimolecular reaction is favoured on large pore zeolites, like HY. Still, the higher occurrence of oligomerization during PP conversion would suggest that primary products from this feedstock would be more likely to undergo oligomerization, a phenomenon possibly linked to the higher branching of PP. Finally, another critical factor for the performance of zeolite catalysts in the catalytic cracking of hydrocarbons is coke formation. Indeed, Brillis and Manos showed that isomer hydrocarbons promote faster deactivation than linear equivalents during catalytic cracking on Y zeolites.[46,51] Given this, coke selectivity for both feedstocks was similar at 1.8 wt%, showing no particular indication of preferred coke formation.

Unlike HY and HZSM-5, HBEA displayed higher selectivity to light products when converting PP compared to LDPE, *i.e.*, a gas yield of 47.5 wt% and 35.3 wt%, respectively. On the one hand, like HY, the large pores and 3D framework configuration of HBEA allows the diffusion of bulky molecules without shape selectivity. However, this zeolite's small crystallites and large mesopore volume (Table 1) can significantly reduce diffusion limitations, contributing to a more effective conversion of the more reactive PP. On the other hand, HBEA displayed high Brønsted and Lewis acidity (Table 1), further enhancing the cracking of LDPE, PP, and respective intermediates into lighter products, *i.e.*, gas products.

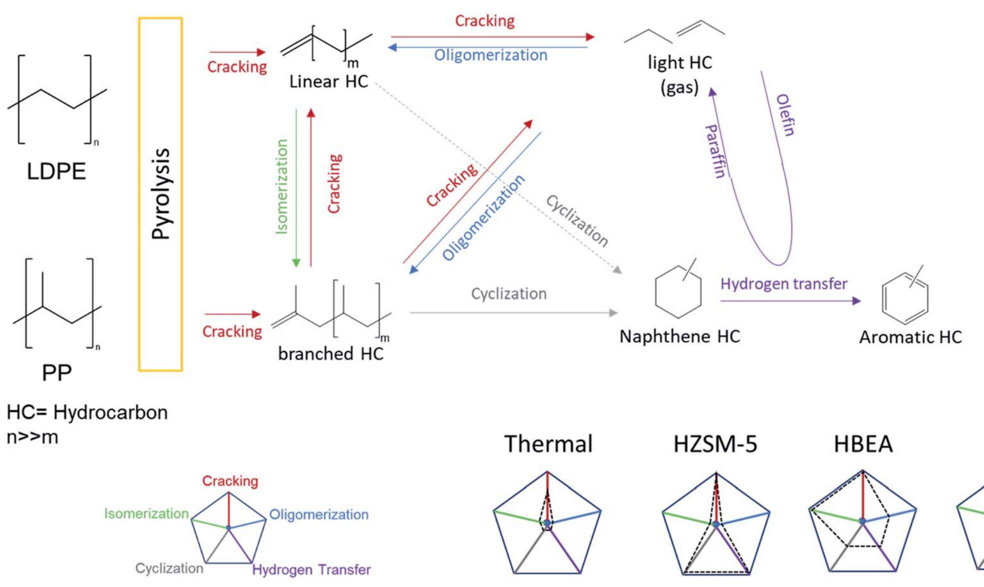
The complexity of the reaction pathways occurring during the conversion of LDPE and PP over the different zeolites significantly limits the comprehension of the phenomena through the simple analysis of gas, liquid, and coke yield. Consequently, the detailed analysis of each fraction composition and the physicochemical properties of the spent catalysts will be discussed in the following sections to provide clear insights into the transformation mechanisms of LDPE and PP.

3.2.2. Gas products. The light gas product yields obtained from the catalytic and thermal conversion of LDPE and PP are reported in Table 2. Despite the similar gas yields from PP and LDPE under thermal conditions, the composition of the gas products was significantly different. LDPE displayed similar C_2 and C_3 hydrocarbons yields, *i.e.*, 5.7 wt% and 5.6 wt%, respectively, with an even selectivity between olefin and paraffin. By contrast, PP was particularly selective to C_3 hydrocarbons, especially propylene, with a yield of 10.8 wt%. The difference in gas product selectivity between PP and LDPE is linked to the different structures of the polymers. Due to the linear hydrocarbon structure, LDPE undergoes thermal cracking randomly, which leads to an even distribution of the products by carbon number.[52] By contrast, PP's highly branched structure includes tertiary carbons, which generate more stable radical intermediates and can lead to the preferential formation of propylene.[53]

In the presence of the zeolite catalysts, the conversion of PP and LDPE occurs primarily through an acid-promoted reaction instead of a radical mechanism. The capacity of the zeolite active sites to promote catalytic cracking of LDPE, PP, and their derived products, explains the higher yield of gas when these were present in the reaction. The higher selectivity towards C_3 hydrocarbons, in particular, during the catalytic pyrolysis of LDPE, is also explained by the acid cracking mechanism. At the reaction temperature, *i.e.*, 450 °C, catalytic cracking of hydrocarbons over acid zeolites occurs mainly through β-scission, which does not favour the formation of C_2 hydrocarbons, as these occur *via* an unstable primary carbocation intermediate.[54]

Table 2 Yields of light gas products

		Gas yield (wt%)	Methane (wt%)	Ethylene (wt%)	Ethane (wt%)	Propylene (wt%)	Propane (wt%)	Propylene/Propane
LDPE	Thermal	15.6	1.3	2.4	3.1	2.9	2.7	1.1
	HY	36.1	0.3	0.6	0.5	9.2	1.4	6.4
	HBEA	35.3	0.1	0.4	0.2	11.1	0.6	18.1
	HZSM-5	68.7	0.2	6.7	0.3	24.0	11.1	2.2
PP	Thermal	18.9	1.2	0.9	2.6	10.8	0.4	29.5
	HY	19.4	1.2	1.0	0.8	4.8	0.8	5.7
	HBEA	47.5	1.3	1.2	0.03	12.8	0.6	21.8
	HZSM-5	51.5	0.3	3.9	0.6	12.1	13.9	0.9



Scheme 1 Reaction mechanism for the formation of liquid and gas products during the pyrolysis of LDPE and PP under thermal conditions and promoted by HZSM-5, HBEA, and HY zeolites.

In addition to promoting cracking, the acid sites of zeolites promote hydrogen transfer and condensation reactions. Condensation reactions can also explain the lower tendency of HY zeolite to yield propylene during PP conversion compared to thermal pyrolysis. Indeed, the sodalite cages of HY can easily accommodate large molecules (Table 1), particularly when compared with HZSM-5, where the catalyst's framework and shape selectivity hinder condensation reactions.[54] On the other hand, hydrogen transfer can explain the different ratios between propylene and propane observed over the different catalysts. Indeed, hydrogen transfer leads to a higher occurrence of saturated hydrocarbons, such as propane, and aromatic compounds[54] (Scheme 1). Hence, the lower ratio of $\frac{\text{propylene}}{\text{propane}}$ observed when using HZSM-5 zeolite (Table 2), despite the high yield of propylene, as a direct consequence of hydrogen transfer. Still, HZSM-5 displayed the highest selectivity towards light olefins from the three tested zeolites, with the highest selectivity being attained on LDPE, with ethylene and propylene yields of 6.7 wt% and 24.0 wt%, respectively. It is important to mention that condensation and hydrogen transfer reactions are favoured in samples with higher density of Brønsted acid sites,[55–57] like HZSM-5 and HBEA. Yet, HBEA displayed the highest $\frac{\text{propylene}}{\text{propane}}$ ratio suggesting minimal hydrogen transfer occurrence during the reaction. Consequently, while parent HBEA has high acid site density, the catalyst responsible for the main conversion of the polymers and derived products does not. The partial deactivation, by coke deposition, of the zeolite's acid sites during the first moments of the reaction could explain the observed phenomenon. While the discussion on coke products and catalysts deactivation will be done below, it is worth mentioning that Câmara *et al.* demonstrated HBEA zeolite to undergo coking significantly faster HZSM-5 and HBEA,[58] supporting the previously stated hypothesis.

3.2.3. Liquid products. The carbon number distribution of the pyrolysis volatiles captured in the condenser is presented in Fig. 3. The liquid product distributions obtained from thermal pyrolysis of LDPE and PP were significantly different. Despite the similar liquid yield (Fig. 2), the thermal pyrolysis of PP produced more of the gasoline range hydrocarbons than that of PE, *i.e.*, 72% *vs.* 44% of liquid products under C_{12}. The liquid obtained from LDPE thermal pyrolysis showed an even distribution, which is a consequence of the linear hydrocarbon structure of LDPE undergoing thermal cracking randomly,[52] as previously discussed. On the other hand, PP thermal pyrolysis led to the preferential formation of C_9, C_{11}, and C_{14} hydrocarbons (Fig. 3). The higher incidence of specific carbon number products during thermal pyrolysis of PP derives from the methyl groups attached to the main polymer chain, which leads to preferential cracking. Kiang *et al.* studied the mechanism behind the thermal degradation of PP, identifying the formation of C_9 as the main occurring product.[59]

The presence of the catalyst significantly changed the carbon distribution of the liquid products compared to non-catalytic or thermal pyrolysis. In addition, some minimal differences in carbon distribution of liquid products can be seen between LDPE and PP (Fig. 3). The main products are in the gasoline range, *i.e.*, from C_6 to C_{12}, independent of the zeolite used. Indeed, the liquid products from LDPE conversion in the presence of HY, HBEA, and HZSM-5 contained 85 wt%, 91 wt%,

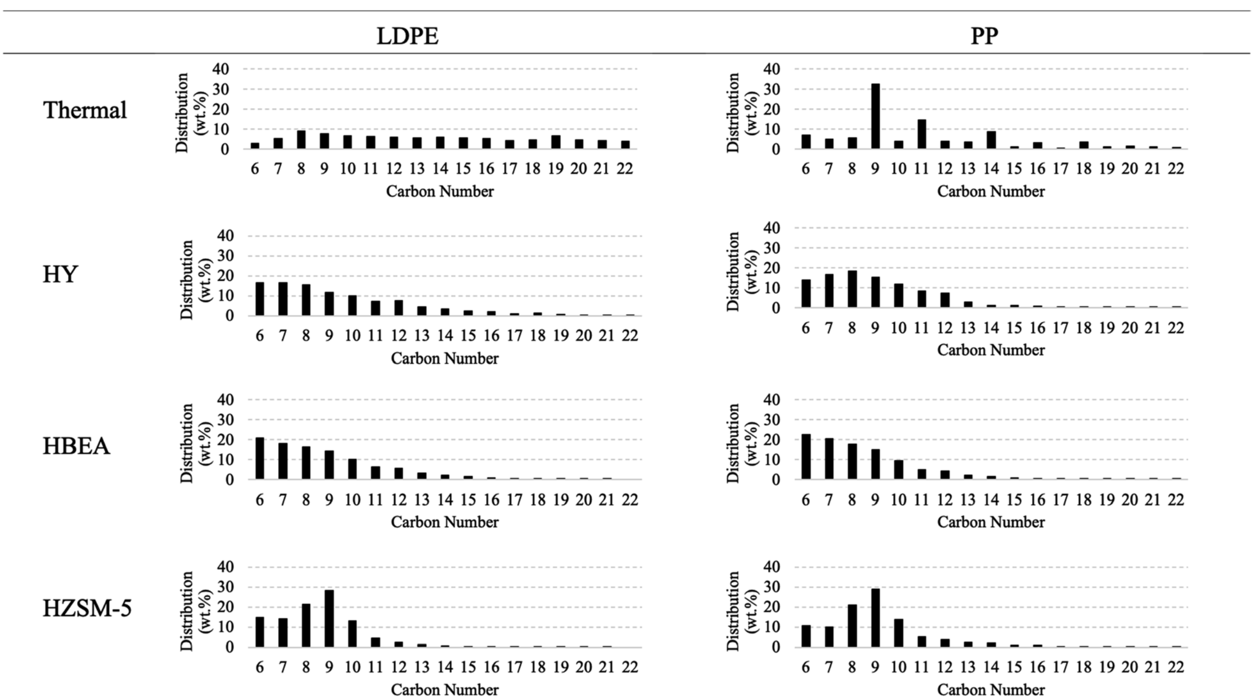


Fig. 3 Carbon distribution of liquid products from thermal and catalytic pyrolysis of LDPE and PP.

and 98 wt% of C_6–C_{12} hydrocarbons, respectively. On the other hand, the catalytic pyrolysis of PP yielded a liquid product with 92 wt%, 94 wt%, and 93 wt% of C_6–C_{12} hydrocarbons over HY, HBEA, and HZSM-5, respectively. The significant selectivity towards gasoline range products for all catalysts and both polymers indicates that secondary cracking reactions occur in the stirred tank reactor, even at such low temperatures as 450 °C. In fact, Elordi *et al.* studied the conversion of HDPE over HBEA, HZSM-5, and HY zeolites at 500 °C in a spouted bed reactor, observing a much higher selectivity, *i.e.*,>10 wt%, of C_{12}^+ and wax products, in particular over HY.[21] It is important to note that the spouted bed reactor is known for a low residence time of volatiles, *i.e.*, yielding low or minimal secondary reactions.

Even though most liquid products occur in the same carbon range, some differences caused by zeolite frameworks are evident (Fig. 3). HZSM-5 liquid products display the highest selectivity towards C_9 hydrocarbons, whereas HY favoured C_7–C_8 and HBEA towards C_6. It is important to mention that even if the liquid products obtained with the different zeolites and feedstocks are similar in terms of carbon distribution, the overall liquid yield changed significantly (Fig. 2). Hence, HY is the catalyst with the highest yield towards gasoline range products, *i.e.*, 53 wt% and 72 wt% yield of C_6–C_{12} products from LDPE and PP. By contrast, HZSM-5 led to the lowest C_6–C_{12} products yield from LDPE and PP, *i.e.*, 30 wt% and 45 wt%, respectively. The high capacity of HY zeolite to yield gasoline range products from the high molecular weight hydrocarbons is not surprising since this catalyst is the main active phase used in FCC catalysts for this same purpose.[60] Similarly, HZSM-5 is a common additive to FCC catalysts to increase the yield of light olefins, particularly propylene, at the expense of gasoline range products.[60,61]

Cracking is not the only reaction responsible for forming liquid range products. Indeed, the same molecules can also be obtained through the oligomerization of small olefins. Additionally, monoaromatic compounds resultant from hydrogen transfer are also part of the liquid products with C_6–C_{12}. In the previous section, the analysis of the gas products suggested that HZSM-5 and HY were particularly prone to promoting hydrogen transfer and the condensation of small olefins, respectively. Therefore, the higher selectivity of HZSM-5 and HY towards C_7–C_9 products might result from secondary reactions.

The distribution of types of carbon-hydrogen (C–H) bonds in liquid products, which could help understand the conversion pathways, are shown in Fig. 4. After thermal pyrolysis, the products for PP and LDPE were mainly composed of aliphatic C–H bonds, at ~84%. Still, the aliphatic nature of the products was significantly different between LDPE and PP, where the products from the thermal pyrolysis of LDPE contained 67% of aliphatic CH_2, while PP had 17%. Additionally, the LDPE liquid contained 15% of aliphatic CH_3 bonds and 3.5% of aliphatic CH, whereas PP liquid was composed of 50% of aliphatic CH_3 and 16% of aliphatic CH. The significant differences observed in the composition of liquid products from the thermal pyrolysis of LDPE and PP reflect the differences in the chemical structures of the two polymers. It is important to mention that during thermal pyrolysis isomerization, hydrogen transfer, cyclization, and aromatization reactions also occur in addition to thermal cracking. While cyclization and aromatization reactions are responsible for the formation of aromatics in the liquid (Fig. 4), isomerization and transfer hydrogenation lead to the formation of CH and CH_3 bonds, respectively.[62] Hence, justifying the occurrence of such bonds on the LDPE thermal cracking product.

The presence of a zeolite significantly changed the aliphatic structure of the liquid products obtained from the pyrolysis of LDPE and PP (Fig. 4). For LDPE, the proportion of aliphatic CH_2 bonds significantly decreased with the appearance of more CH and CH_3 due to cracking, hydrogen transfer, and isomerization reactions (Scheme 1). The bonds of carbon linked to a single hydrogen (aliphatic CH), in particular, can be used to indicate the degree of branching of the aliphatic carbon of the liquid products. Hence, the occurrence of isomerization reactions under different pyrolysis conditions can be investigated by comparing the proportion of aliphatic CH in the liquid to other forms of aliphatic CH bonds ($\frac{\text{aliphatic } \mathbf{CH}}{\text{aliphatic } \mathbf{CH_n}}$, Table 3).

For LDPE, the $\frac{\text{aliphatic } \mathbf{CH}}{\text{aliphatic } \mathbf{CH_n}}$ ratio increased from 0.04 for thermal pyrolysis to 0.13, 0.23, and 0.13 with the use of HY, HBEA, and HZSM-5, respectively. The higher capacity of HBEA for the isomerization of linear hydrocarbons, when compared to other zeolites, has been reported previously under hydroisomerization conditions and is associated with the particular intermediate size framework structure (Table 1) and milder acid

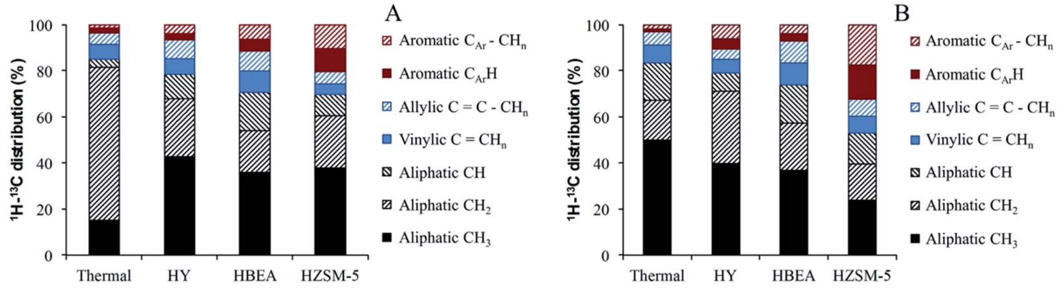


Fig. 4 The carbon–hydrogen bond type distribution in liquid products, determined by ^{1}H–^{13}C 2D NMR (HSQC), obtained from the pyrolysis of LDPE (A) and PP (B). NMR spectra in the ESI.†

Table 3 Relationship between different bonds abundance in the liquid products (bond distribution in Fig. 4)

		$\frac{C_{Ar}-\mathbf{CH}_n}{\mathbf{C_{Ar}H}}$ [a]	$\frac{Ca\mathbf{CH}_n}{CaC-\mathbf{CH}_n}$ [b]	$\frac{\text{Aliphatic }\mathbf{CH}}{\text{Aliphatic }\mathbf{CH}_n}$ [c]
LDPE	Thermal	0.6	1.3	0.04
	HY	1.5	0.8	0.13
	HBEA	1.2	1.0	0.23
	HZSM-5	1.1	0.9	0.13
PP	Thermal	1.3	1.4	0.19
	HY	1.3	1.3	0.10
	HBEA	1.1	1.0	0.22
	HZSM-5	1.1	0.9	0.13

[a] Ratio between alkyl aromatic bonds (C_{Ar}–$\mathbf{CH}_n$) and non-alkyl aromatic bonds ($\mathbf{C_{Ar}H}$). Defines the alkylation level of aromatic products. [b] Ratio between vinylic (C=$\mathbf{CH}_n$) and allylic (C=C–$\mathbf{CH}_n$) bonds. Defines the relative proportion of α-olefins among samples. [c] Proportion of aliphatic **CH** bonds in among all aliphatic **CH** bonds. Ratio indicative of isomerization level of products.

strength of this zeolite.[63–65] On the other hand, HZSM-5 and HY zeolites were found to be more suited for cracking reactions.[64,65] Indeed, the shape selectivity of HZSM-5 hinders the diffusion of highly isomerized products, which undergo cracking inside the micropores of the zeolite. By contrast, the much larger pores of HY readily permit isomerization of the linear hydrocarbons into multi-branched isomers, which can easily undergo cracking.[64] Therefore, any branched products formed in the presence of HY and HZSM-5 are more likely to undergo cracking, decreasing the $\frac{\text{aliphatic }\mathbf{CH}}{\text{aliphatic }\mathbf{CH}_n}$ ratio. The higher selectivity towards cracking over isomerization of HZSM-5 and HY[64,65] helps to explain the decrease in the liquid $\frac{\text{aliphatic }\mathbf{CH}}{\text{aliphatic }\mathbf{CH}_n}$ ratio observed when these zeolites were used in the pyrolysis of PP.

Aromatic compounds are the product of the cyclization followed by hydrogen transfer of olefins (Scheme 1). The analysis of gas products distribution suggested the occurrence of hydrogen transfer reactions promoted by the zeolite catalysts. Therefore, analyzing the concentration of aromatic bonds ($C_{Ar}H$) in the liquid will help to verify the hypothesis proposed earlier.

Independently of the zeolite used to promote the reaction, the concentration of aromatic CH bonds ($C_{Ar}H$) increased. HZSM-5 led to the highest concentration of aromatic products with PP liquid containing 15% of $C_{Ar}H$ and 10% for LDPE liquid (Fig. 4). The high concentration of aromatics when using HZSM-5 indicates a high occurrence of condensation and hydrogen transfer reactions, which is in agreement with the low $\frac{\text{propylene}}{\text{propane}}$ ratio observed in the gas products (Table 2) and the high selectivity towards C_7–C_{10} products. The remaining two zeolites (HBEA and HY) were less selective towards forming aromatic products in the liquid, in general being <5% for both polymers. The higher concentration of aromatics in HZSM-5 promoted liquid can be explained by its high acidity and particular shape-selective porous framework, limiting the formation of poly-aromatic compounds, delaying deactivation, and increasing liquid range aromatic products. Indeed, HZSM-5 has been reported multiple times as an efficient catalyst for converting olefins and alkanes into aromatic hydrocarbons.[66–68]

The aromatic products were found to contain a significant degree of alkylation, as confirmed by the presence of CH bonds linked to the aromatic carbons (C_{Ar}–CH_n). The degree of alkylation of the aromatic compounds can be estimated by the $\frac{C_{Ar}-CH_n}{C_{Ar}H}$ ratio, which defines the number of alkylated per non-alkylated aromatic carbons (Table 3). After thermal pyrolysis, the $\frac{C_{Ar}-CH_n}{C_{Ar}H}$ for LDPE and PP were 0.6 and 1.3, respectively – in line with expectations, since one in every two carbons in the PP polymer chain contains a methyl group.

For LDPE, the $\frac{C_{Ar}-CH_n}{C_{Ar}H}$ ratio (Table 3) significantly increased from 0.6, under thermal conditions, to 1.1–1.5 in the presence of zeolite catalysts. The largest degree of alkylation, with $\frac{C_{Ar}-CH_n}{C_{Ar}H}=1.5$, occurred in the presence of HY due to its large pore framework (Table 1), followed by HBEA (1.2) and HZSM-5 (1.1). On the other hand, the $\frac{C_{Ar}-CH_n}{C_{Ar}H}$ ratio in aromatics obtained from PP was found similar for thermal and HY promoted reactions but decreased to 1.1 for both HBEA and HZSM-5. The lower alkylation level of the aromatic products from PP in the presence of zeolites indicates that a significant portion of the aromatic products was formed from the condensation of smaller products, *i.e.*, light olefins. Such a phenomenon also explains the small increase of propylene yield over HBEA and HZSM-5, or decrease in the case of HY, in comparison to thermal cracking of PP, despite the significant increase in overall gas yield (Table 2). Additionally, due to the shape selectivity effect, the HZSM-5 framework does not allow the diffusion of highly alkylated molecules from PP conversion. Therefore, the reconversion of small gas molecules could explain the high presence of aromatics in the liquid when using HZSM-5. The same mechanism is responsible for the formation of aromatic products during catalytic pyrolysis of biomass.[69,70] Yet, independently of the feedstock, it is clear that the determining fact contributing to the number of alkyl groups in the aromatic products is the zeolite framework pore size, *i.e.*, HY > HBEA > HZSM-5 (Table 1).

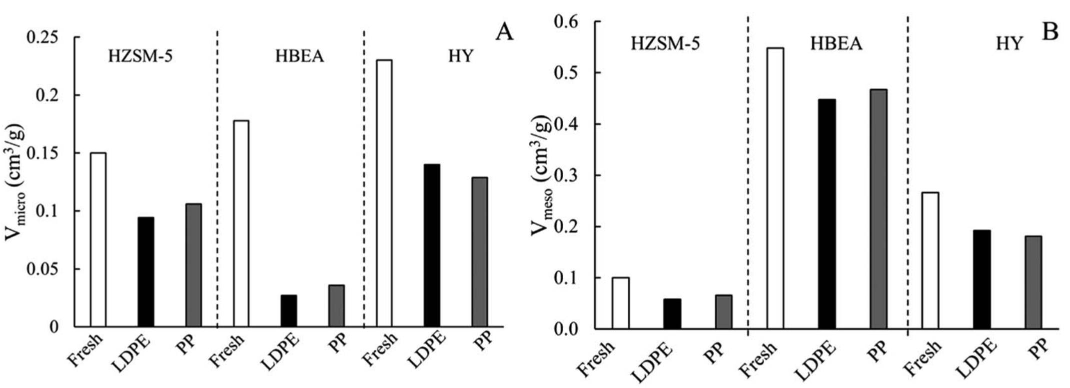
In order to produce liquid and gas products, the long polymer chains of LDPE and PP undergo multiple cracking steps, generating a significant number of α-olefins.[71] The olefins generated during the primary cracking steps can undergo, *via* a radical or acid mechanism, oligomerization, isomerization alkylation, cyclization, and hydrogen transfer to yield products other than α-olefins, *i.e.*, non-α-olefins, naphthenes, paraffin, and aromatics (Scheme 1). The ratio of $\frac{Ca\mathbf{CH}_n}{CaC-\mathbf{CH}_n}$ relates to the number of vinylic CH bonds to the number of allylic CH and is equal to 1.5 for α-olefins, without alkyl groups in allyl position. Thus, the closer the ratio is to 1.5 the higher the proportion of α-olefins in the liquid relative to the total number of olefins.

Under thermal conditions, the $\frac{\mathrm{Ca}\mathbf{CH}_n}{\mathrm{CaC}-\mathbf{CH}_n}$ was 1.36 and 1.40 for LDPE and PP, respectively. Indeed, LDPE and PP displayed similar pyrolysis profiles under thermal conditions (Fig. 2), due to the similar C–C bond energy of the two polymers.[15] On the other hand, the ratio of $\frac{\mathrm{Ca}\mathbf{CH}_n}{\mathrm{CaC}-\mathbf{CH}_n}$ in presence of zeolites were always smaller than in the thermal conditions (Table 3), despite the liquid products from catalytic pyrolysis having carbon distributions in the lower range (Fig. 3). As already noted, the acid sites of zeolites promote oligomerization and isomerization reactions, which can easily convert α-olefins into standard olefins. Such a phenomenon is stronger during the conversion of LDPE, where the $\frac{\mathrm{Ca}\mathbf{CH}_n}{\mathrm{CaC}-\mathbf{CH}_n}$ stayed between 0.8 and 1.0, *versus* PP, where the ratio of was in between 1.3 and 0.9. The higher $\frac{\mathrm{Ca}\mathbf{CH}_n}{\mathrm{CaC}-\mathbf{CH}_n}$ observed during the catalytic pyrolysis of PP might be due to the higher degree of isomerization in PP and its primary cracking products, which significantly decreases the activation energy during cracking reactions by favouring the occurrence of tertiary carbocations.[45] It is important to mention that this effect should be lessened on HZSM-5, due to steric limitations (shape selectivity), and, for this reason, these samples displayed a similar $\frac{\mathrm{Ca}\mathbf{CH}_n}{\mathrm{CaC}-\mathbf{CH}_n}$ ratio of 0.9 for the conversion of LDPE and PP.

The reaction diagram shown in Scheme 1 highlights the main reactions through which LDPE and PP are transformed during pyrolysis. In addition to enhancing cracking, the analysis of liquid and gas products revealed that secondary reactions on the zeolite catalysts significantly impact the distribution of the products. Conversely, reactions such as isomerization, cyclization, and hydrogen transfer are minimized under thermal conditions, with cracking being the primary reaction occurring. Among the three tested zeolites, ZSM-5 displayed significantly higher selectivity towards products from cyclization/hydrogen transfer, *i.e.*, aromatics and light paraffins. On the other hand, HBEA was more selective towards isomerization and HY towards oligomerization. Hydrogen transfer and cyclization products were also observed on HY and HBEA, however to a smaller extent than on HZSM-5.

3.2.4. Coke products and catalyst deactivation. In addition to the gas and liquid products, the catalytic pyrolysis of the LDPE and PP also led to the formation of heavier products, also known as coke, which remain on the catalyst after reaction and ultimately lead to its deactivation. The micropore (V_{micro} – diameter < 2 nm) and mesopore volume (V_{meso} – 2 < diameter < 50 nm) before (fresh) and after reaction, the coke elemental composition, the loading of coke, and residual acidity of the spent catalysts are shown in Fig. 5. In general, the amount of coke deposited on the samples and the losses observed in the textural properties were mainly defined by the zeolite's porous structure, *e.g.*, pore size and pore interconnectivity,[72] with the type of feedstock having minimal impact on these parameters. HY retained the highest coke loading, at ∼15.5 wt%, followed by HBEA, ∼12 wt%, and HZSM-5, ∼3.6 wt%. Coke retention correlated with the available accessible volume inside each zeolite framework, *i.e.*, HY (27.4%) > HBEA (20.5%) ≫ HZSM-5 (9.8%).[41]

The presence of coke in the zeolite led to a decrease of the available pore volume (V_{micro}), which corresponds to the space inside the zeolite crystalline framework (Fig. 5A), being significantly reduced. HZSM-5 lost 37% and 29% of its V_{micro} after reaction with LDPE and PP, respectively, despite the smaller pore structure. Indeed, the low coke formation properties and resistance towards deactivation by coke for HZSM-5 zeolite have



C

	LDPE			PP		
	Coke loading (wt.%)	H/C (molar)	Brønsted sites (μmol/mol)	Coke loading (wt.%)	H/C (molar)	Brønsted sites (μmol/mol)
HY	15.4	0.9	16 (-94%)	15.5	0.8	33 (-88%)
HBEA	12.4	1.0	106 (-78%)	11.8	1.0	92 (-81%)
HZSM-5	3.5	1.2	260 (-48%)	3.7	1.1	306 (-40%)

Fig. 5 Micropore (A) and mesopore (B) volume of fresh and spent catalysts. (C) Coke loading (wt% – dry basis), H/C ratio of coke components on spent catalysts after cracking reaction, and spent catalyst Brønsted acidity (values in brackets represent acidity lost during reaction).

been consistently reported and attributed to the particular microporous structure of this zeolite, limiting the formation of coke precursors.[16,43,73,74] The resistance to deactivation of HZSM-5 zeolite is also reflected in the spent catalyst's residual acidity, with 60% and 52% of the Brønsted acid sites of the zeolite still being accessible to pyridine after reaction with PP and LDPE, respectively (Fig. 5C). On the other hand, HBEA lost 85% and 80% of its V_{micro} after reaction with LDPE and PP, together with ~80% of its Brønsted acidity, indicating this zeolite suffered severe deactivation during the reaction. Finally, HY only lost 34% and 39% of V_{micro} due to coke deposition after reaction with LDPE and PP, respectively. Despite the relatively small loss of V_{micro}, HY lost 88–94% of the zeolite's Brønsted acidity, indicating the coke was selectively poisoning the zeolites acid sites without completely blocking the microporous framework. Indeed, pyridine's kinetic diameter (0.533 nm) is significantly bigger than N_2 (0.364 nm), thus nitrogen diffused inside the zeolite more easily despite partial blocage by coke. It is important to mention that LDPE and PP pyrolysis products are also bigger than N_2, hence their accessibility to the acid sites, much like that of pyridine, must also be hindered on spent HY. The significant acidity loss observed over HY, when compared HBEA and HZSM-5, contradicts the observations of Elordi *et al.* for the catalytic pyrolysis of HDPE, where after 15 h HY still retained ~40% of its acidity.[16] Still, the authors used NH_3 temperature programmed desorption to assess the loss of acidity and NH_3 has a relatively small kinetic diameter of 0.260 nm, potentially explaining the different conclusions. The hypothesis concerning the partial blockage of HY pores by coke molecules is supported by the average size of coke molecules trapped inside the zeolite pores. As will be discussed below, the coke trapped inside HY pore has, on average, 3.0 aromatic cycles for LDPE and 3.4 for PP. Consequently, a direct comparison between the kinetic diameter of anthracene (0.696 nm)/pyrene (0.724 nm)[75] and the size of HY size of the largest sphere, which can be included on HY(1.124 nm) reveals an empty space must exist.

The selective deposition of coke inside HY zeolite without causing complete blockage could help explain the unchanged mesopore volume (V_{meso}) after reaction with LDPE and PP (Fig. 5B). Indeed, after extraction with CH_2Cl_2 at 110 °C, only 0.5% of the coke mass, corresponding to the external coke, was extracted. On the other hand, HZSM-5 lost 42% and 34% of V_{meso} after reaction with LDPE and PP, while V_{meso} of HBEA was reduced by 18% and 14%, respectively. After CH_2Cl_2 extraction, HZSM-5 and HBEA external coke was 10% and 5% of their total coke loading for LDPE and PP, respectively. The small loss of V_{meso} suggests that deposition of unconverted polymer and coke on the surface of the catalysts is minimal. Indeed, GC/MS analysis of the extracted coke from the spent catalyst surface (not shown) showed this fraction was composed mainly of aromatic compounds. This observation is also supported by the low H/C ratio of the coke products (Fig. 5C), which stayed between 0.8 and 1.2 for all catalysts, indicating coke is mainly composed of aromatic molecules.

The composition of the soluble coke is shown in Fig. 6. It is important to mention that the amount of insoluble coke in CH_2Cl_2 after zeolite mineralization was minimal, *i.e.*, <5%, and for this reason, not analyzed. The lack of insoluble coke can be explained by the low reaction temperature and short reaction time, *i.e.*, 450 °C and 30 min, which do not favour the formation of more refractory coke molecules.[72] Independently of the zeolite structure and feedstock, soluble coke was exclusively composed of aromatic hydrocarbons, as suggested by the low H/C ratio shown in Fig. 5C. The distribution of the aromatic compounds depends on both the zeolite framework and the type of feedstock. HZSM-5 coke generated during the conversion of LDPE contained on average 3.1 aromatic cycles against 2.6 when PP was used as feedstock. On the other hand, the opposite behaviour was observed for HY and HBEA, in which coke was composed of aromatics with an average of 3.0 and 3.1 cycles, respectively, for LDPE and 3.4 cycles for PP.

The soluble coke corresponds to the molecules trapped inside the zeolite framework, which, much like liquid range

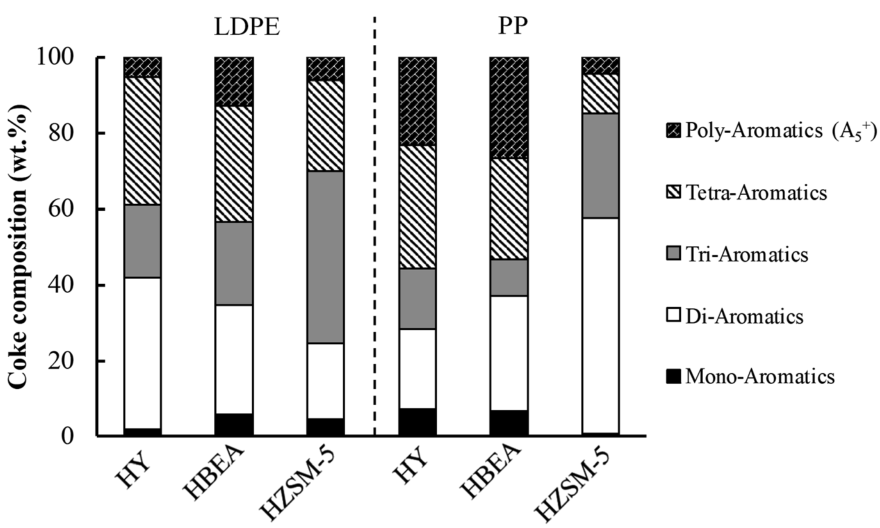


Fig. 6 Distribution of products in coke soluble in CH_2Cl_2, often referred as soluble coke, after zeolite mineralization.

products, are subjected to the shape selectivity effect of the zeolite framework. Therefore, the small pores of the HZSM-5 framework can significantly impact the nature of the poly-aromatic molecules composing coke. Mihindou-Koumba *et al.* studied coke formation kinetics on HZSM-5, HBEA, and HY zeolites during methylcyclohexane conversion, verifying that HZSM-5 soluble coke was composed of smaller molecules with a smaller degree of aromaticity than HY and HBEA.[76] Much like the number of aromatic rings, alkyl groups can also significantly increase the size of a molecule, *e.g.*, the kinetic diameter of naphthalene and dimethylnaphthalene are 0.62 nm and 0.72–0.77 nm,[75] respectively. The average number of alkyl groups of soluble coke in HZSM-5 after catalytic pyrolysis of PP was 3.7, while for LDPE, this value was 2.6. Thus, the higher isomerization degree of PP and its derived products generated a coke with a higher degree of alkylation, which limited further aromatization. On HBEA and HY, where the zeolite framework has larger pores, the degree of alkylation was similar for LDPE and PP and equal to 2.9 and 3.1, respectively. Additionally, the absence of severe shape selectivity limitations for the more reactive intermediates resulting from PP pyrolysis led to a higher degree of coke aromatization on HBEA and HY samples, indicating PP can lead to a faster deactivation of these catalysts than LDPE.

The analysis of the physicochemical properties of the spent catalyst and the composition of the coke responsible for the deactivation of the catalysts showed significantly different deactivation behaviour among the tested zeolites. The similar loss of V_{micro} and Brønsted acidity indicates deactivation on these zeolites occurs mainly by pore blockage, as the size of the coke molecules found on these zeolites is similar to the framework pore size. However, without surprise, the deactivation was significantly slower on HZSM-5. On the other hand, on HY deactivation occurs mainly through poisoning of the active sites by coke, as demonstrated by the relatively high pore volume of the spent catalyst, small size of coke molecules *versus* framework size, and significant loss of acidity.

3.3. Outlook: catalytic pyrolysis in a stirred tank reactor (STR)

The catalytic pyrolysis of the LDPE and PP was performed in the presence of HBEA, HZSM-5, and HY zeolites in a stirred tank reactor with continuous removal of volatiles, by opposition to the more conventional fluidized bed reactor. Taking the fluid catalytic cracking (FCC) process as an example, fluidized bed reactor systems have very short product residence times (~2 s) inside the reactor and use an average catalyst to feedstock ratio of 5.5, which leads to an average contact time of 11 s kg_{cat} $kg_{feedstock}^{-1}$.[60] In this work, we used a catalyst to feedstock ratio of 0.1 with an average product residence time was 200 s $\left(\mathrm{Prod}_{\mathrm{residence\ \ time}} = \dfrac{V_{\mathrm{reactor}}}{\mathrm{flow}_{N_2}}\right)$, leading to a contact time of 20 s kg_{cat} $kg_{feedstock}^{-1}$. While both values are in the same range, the nature plug flow regime of fluidized bed reactor makes all products exit the reactor simultaneously. By opposition, only volatile products leave the reactor on the stirred tank reactor (boiling point n-C_{30} = 451 °C), making the heavier compounds stay inside the reactor and in contact with the catalyst. This specific behaviour could explain why no wax, *i.e.*, long carbon chain paraffin, was found in the STR after adding zeolite, while such products are commonly obtained with fluidized bed or spout bed reactors.[16,21] Similarly, the longer residence time of the feedstock and heavier intermediates in contact with the catalyst, resultant from the nature of the operation of STR, can explain the higher gas products yield, particularly over the zeolites with higher acidity and resistance to deactivation, *i.e.*, HZSM-5. It should be noted that the product selectivity can be tuned by modifying the catalyst structure, acidity, and amount, as well as the reaction operating conditions, *e.g.*, temperature and reactor type.

In addition to the differences in the residence time distribution, the amount of catalyst used feedstock in the STR reactor was significantly smaller, *i.e.*, 0.1, than the average 5.5 practiced in the FCC process[60] (example for fluidized bed). Despite the lower catalyst to ratio, significant polymer conversion was achieved, indicating the catalyst could still promote the reaction without undergoing complete deactivation in the initial moments of the reaction. Indeed, Fig. 5 shows that HZSM-5 and HBEA still maintain part of their acidity accessible for reaction after reaction. The significant number of olefins in the reaction media, when compared to the standard refinery charges processed by FCC, resulting from the cleavage of the polymer chain, can explain the considerable activity of the zeolites even at a low reaction temperature, *i.e.*, 450 °C. Indeed, the rate-limiting step of paraffin cracking is the initial activation of the paraffin by the acid sites (monomolecular cracking), which olefins do not require to undergo cracking.[77,78] Compared to the fluidized bed, the requirement for lower catalyst usage by the STR system can be seen as a significant advantage of this system since catalyst makeup accounts for a considerable part of FCC operating cost.[60]

Finally, the study on the catalytic pyrolysis of LDPE and PP on STR reactor showed that the addition of small amounts of zeolite catalysts to reactor systems, which have a long feedstock residence time, *e.g.*, CSTR, auger, rotating kiln, *etc.*, can significantly enhance the conversion of the polymer into gas and liquid range products, even at relatively low temperature (450 °C). Consequently, these simple and inexpensive systems, *i.e.*, when compared to fluidized bed reactors, could provide a suitable alternative for the valorization of plastic waste.

4. Conclusion

Under thermal and catalytic conditions, the pyrolysis of LDPE and PP was performed in a stirred tank reactor. The LDPE and PP conversion analysis showed that the pathways of conversion of these two polymers could significantly affect product selectivity. At 450 °C, thermal pyrolysis yields of liquid, gas, and residue were near-identical for the two feedstocks. Yet, the presence of methyl groups on PP polymer chain significantly increased the selectivity towards propylene and lower molecular weight liquid products. At the same time, LDPE suffered random C–C scission yielding a distribution of products by carbon number. In the absence of a catalyst, the selectivity towards aromatic compounds was small, indicating that secondary reactions were minimal.

In the presence of HBEA, HZSM-5, and HY, the product selectivity from the catalytic pyrolysis of LDPE and PP clearly showed a significant impact on secondary reactions, like cracking reactions, hydrogen transfer, and oligomerization. On HZSM-5, gas products contain near-identical proportions of olefin and paraffin, together with a significant selectivity towards aromatic products suggesting the occurrence of hydrogen transfer reactions. On the other hand, HY and HBEA displayed higher incidence of oligomerization and isomerization, respectively.

The analysis of the physicochemical properties of the spent catalysts clearly showed HZSM-5 resistance to deactivation, in particular when compared to HY and HBEA, which lost more than 80% of the Brønsted acid sites accessible to pyridine. Additionally, descrepancies between loss of micropore volume and acidity on HY indicated this zeolite deactivation occurs through the poisoning of the active sites by coke molecules rather than by pore blockage, which is the case for HBEA and HZSM-5.

Yet, independently of the catalyst used, the larger residence time permitted the complete conversion of the LDPE and PP polymers, even at relatively low temperature, *i.e.*, 450 °C. Thus, the use of such reactor configurations, which permit a longer residence time of polymers in contact with the catalyst, might open the path for low-temperature catalytic pyrolysis of polyolefins.

Conflicts of interest

There are no conflicts to declare.

References

1 L. S. Dilkes-Hoffman, S. Pratt, B. Laycock, P. Ashworth and P. A. Lant, Public Attitudes towards Plastics, *Resour., Conserv. Recycl.*, 2019, **147**, 227–235, DOI: 10.1016/j.resconrec.2019.05.005.

2 L. Lebreton and A. Andrady, Future Scenarios of Global Plastic Waste Generation and Disposal, *Palgrave Commun.*, 2019, **5**(1), 6, DOI: 10.1057/s41599-018-0212-7.

3 R. Geyer, J. R. Jambeck and K. L. Law, Production, Use, and Fate of All Plastics Ever Made, *Sci. Adv.*, 2017, **3**(7), e1700782, DOI: 10.1126/sciadv.1700782.

4 M. Okan, H. M. Aydin and M. Barsbay, Current Approaches to Waste Polymer Utilization and Minimization: A Review, *J. Chem. Technol. Biotechnol.*, 2019, **94**(1), 8–21, DOI: 10.1002/jctb.5778.

5 S. Budsaereechai, A. J. Hunt and Y. Ngernyen, Catalytic Pyrolysis of Plastic Waste for the Production of Liquid Fuels for Engines, *RSC Adv.*, 2019, **9**(10), 5844–5857, DOI: 10.1039/C8RA10058F.

6 G. Fadillah, I. Fatimah, I. Sahroni, M. M. Musawwa, T. M. I. Mahlia and O. Muraza, Recent Progress in Low-Cost Catalysts for Pyrolysis of Plastic Waste to Fuels, *Catalysts*, 2021, **11**(7), 837, DOI: 10.3390/catal11070837.

7 E. Santos, B. Rijo, F. Lemos and M. A. N. D. A. Lemos, A Catalytic Reactive Distillation Approach to High Density Polyethylene Pyrolysis – Part 1 – Light Olefin Production, *Chem. Eng. J.*, 2019, **378**, 122077, DOI: 10.1016/j.cej.2019.122077.

8 A. Coelho, L. Costa, M. M. Marques, I. M. Fonseca, M. A. N. D. A. Lemos and F. Lemos, The Effect of ZSM-5 Zeolite Acidity on the Catalytic Degradation of High-Density Polyethylene Using Simultaneous DSC/TG Analysis, *Appl. Catal., A*, 2012, **413–414**, 183–191, DOI: 10.1016/j.apcata.2011.11.010.

9 M. Artetxe, G. Lopez, M. Amutio, G. Elordi, J. Bilbao and M. Olazar, Light Olefins from HDPE Cracking in a Two-Step Thermal and Catalytic Process, *Chem. Eng. J.*, 2012, **207–208**, 27–34, DOI: 10.1016/j.cej.2012.06.105.

10 Y.-H. Lin, M.-H. Yang, T.-F. Yeh and M.-D. Ger, Catalytic Degradation of High Density Polyethylene over Mesoporous and Microporous Catalysts in a Fluidised-Bed Reactor, *Polym. Degrad. Stab.*, 2004, **86**(1), 121–128, DOI: 10.1016/j.polymdegradstab.2004.02.015.

11 S. Ali, A. A. Garforth, D. H. Harris, D. J. Rawlence and Y. Uemichi, Polymer Waste Recycling over "Used" Catalysts, *Catal. Today*, 2002, **75**(1), 247–255, DOI: 10.1016/S0920-5861(02)00076-7.

12 G. Elordi, M. Olazar, P. Castaño, M. Artetxe and J. Bilbao, Polyethylene Cracking on a Spent FCC Catalyst in a Conical Spouted Bed, *Ind. Eng. Chem. Res.*, 2012, **51**(43), 14008–14017, DOI: 10.1021/ie3018274.

13 J. Aguado, D. P. Serrano, M. D. Romero and J. M. Escola, Catalytic Conversion of Polyethylene into Fuels over Mesoporous MCM-41, *Chem. Commun.*, 1996, (6), 725–726, DOI: 10.1039/CC9960000725.

14 D. P. Serrano, J. Aguado, J. M. Escola and E. Garagorri, Conversion of Low Density Polyethylene into Petrochemical Feedstocks Using a Continuous Screw Kiln Reactor, *J. Anal. Appl. Pyrolysis*, 2001, **58–59**, 789–801, DOI: 10.1016/S0165-2370(00)00153-4.

15 Q. Zhou, L. Zheng, Y.-Z. Wang, G.-M. Zhao and B. Wang, Catalytic Degradation of Low-Density Polyethylene and Polypropylene Using Modified ZSM-5 Zeolites, *Polym. Degrad. Stab.*, 2004, **84**(3), 493–497, DOI: 10.1016/j.polymdegradstab.2004.01.007.

16 G. Elordi, M. Olazar, G. Lopez, P. Castaño and J. Bilbao, Role of Pore Structure in the Deactivation of Zeolites (HZSM-5, Hβ and HY) by Coke in the Pyrolysis of Polyethylene in a Conical Spouted Bed Reactor, *Appl. Catal., B*, 2011, **102**(1), 224–231, DOI: 10.1016/j.apcatb.2010.12.002.

17 D. Serrano, J. Aguado, J. M. Escola and J. M. Rodríguez, Nanocrystalline ZSM-5: A Highly Active Catalyst for Polyolefin Feedstock Recycling, *Stud. Surf. Sci. Catal.*, 2002, **142**, 77–84.

18 G. Elordi, M. Olazar, G. Lopez, M. Artetxe and J. Bilbao, Continuous Polyolefin Cracking on an HZSM-5 Zeolite Catalyst in a Conical Spouted Bed Reactor, *Ind. Eng. Chem. Res.*, 2011, **50**(10), 6061–6070, DOI: 10.1021/ie2002999.

19 D. P. Serrano, J. Aguado, J. M. Escola and J. M. Rodríguez, Influence of Nanocrystalline HZSM-5 External Surface on the Catalytic Cracking of Polyolefins, *J. Anal. Appl. Pyrolysis*, 2005, **74**(1), 353–360, DOI: 10.1016/j.jaap.2004.11.037.

20 J. Aguado, D. P. Serrano, J. M. Escola, E. Garagorri and J. A. Fernández, Catalytic Conversion of Polyolefins into Fuels over Zeolite Beta, *Polym. Degrad. Stab.*, 2000, **69**(1), 11–16, DOI: 10.1016/S0141-3910(00)00023-9.

21 G. Elordi, M. Olazar, G. Lopez, M. Amutio, M. Artetxe, R. Aguado and J. Bilbao, Catalytic Pyrolysis of HDPE in Continuous Mode over Zeolite Catalysts in a Conical Spouted Bed Reactor, *J. Anal. Appl. Pyrolysis*, 2009, **85**(1), 345–351, DOI: 10.1016/j.jaap.2008.10.015.

22 G. Manos, I. Y. Yusof, N. Papayannakos and N. H. Gangas, Catalytic Cracking of Polyethylene over Clay Catalysts. Comparison with an Ultrastable Y Zeolite, *Ind. Eng. Chem. Res.*, 2001, **40**(10), 2220–2225, DOI: 10.1021/ie001048o.

23 A. Marcilla, M. d. R. Hernández and Á. N. García, Degradation of LDPE/VGO Mixtures to Fuels Using a FCC Equilibrium Catalyst in a Sand Fluidized Bed Reactor, *Appl. Catal., A*, 2008, **341**(1), 181–191, DOI: 10.1016/j.apcata.2008.02.041.

24 E. Rodríguez, A. Gutiérrez, R. Palos, F. J. Vela, M. J. Azkoiti, J. M. Arandes and J. Bilbao, Co-Cracking of High-Density Polyethylene (HDPE) and Vacuum Gasoil (VGO) under Refinery Conditions, *Chem. Eng. J.*, 2020, **382**, 122602, DOI: 10.1016/j.cej.2019.122602.

25 F. J. Vela, R. Palos, J. Bilbao, J. M. Arandes and A. Gutiérrez, Effect of Co-Feeding HDPE on the Product Distribution in the Hydrocracking of VGO, *Catal. Today*, 2020, **353**, 197–203, DOI: 10.1016/j.cattod.2019.07.010.

26 A. O. Odjo, A. N. García and A. Marcilla, Conversion of Low Density Polyethylene into Fuel through Co-Processing with Vacuum Gas Oil in a Fluid Catalytic Cracking Riser Reactor, *Fuel Process. Technol.*, 2013, **113**, 130–140, DOI: 10.1016/j.fuproc.2013.03.008.

27 G. Lopez, M. Artetxe, M. Amutio, J. Bilbao and M. Olazar, Thermochemical Routes for the Valorization of Waste Polyolefinic Plastics to Produce Fuels and Chemicals. A Review, *Renewable Sustainable Energy Rev.*, 2017, **73**, 346–368, DOI: 10.1016/j.rser.2017.01.142.

28 E. Butler, G. Devlin and K. McDonnell, Waste Polyolefins to Liquid Fuels *via* Pyrolysis: Review of Commercial State-of-the-Art and Recent Laboratory Research, *Waste Biomass Valorization*, 2011, **2**(3), 227–255, DOI: 10.1007/s12649-011-9067-5.

29 D. Mcnamara and M. Murray, Conversion of Waste Plastics Material to Fuel, WO2011077419A1, 2011.

30 A. V. Bridgwater, Review of Fast Pyrolysis of Biomass and Product Upgrading, *Biomass Bioenergy*, 2012, **38**, 68–94, DOI: 10.1016/j.biombioe.2011.01.048.

31 J. A. Garcia-Nunez, M. R. Pelaez-Samaniego, M. E. Garcia-Perez, I. Fonts, J. Abrego, R. J. M. Westerhof and M. Garcia-Perez, Historical Developments of Pyrolysis Reactors: A Review, *Energy Fuels*, 2017, **31**(6), 5751–5775, DOI: 10.1021/acs.energyfuels.7b00641.

32 M. Guisnet, P. Ayrault and J. Datka, Acid properties of dealuminated mordenites studied by IR spectroscopy. 2. concentration, acid strength and heterogeneity of OH groups, *Pol. J. Chem.*, 1997, **71**, 1455–1461.

33 P. Magnoux, P. Roger, C. Canaff, V. Fouche, N. S. Gnep and M. Guisnet, New Technique for the Characterization of Carbonaceous Compounds Responsible for Zeolite Deactivation, in *Studies in Surface Science and Catalysis*, ed. B. Delmon and G. F. Froment, Elsevier, 1987, vol. 34, pp. 317–330, DOI: 10.1016/S0167-2991(09)60370-0.

34 K. Möller and T. Bein, Mesoporosity – a New Dimension for Zeolites, *Chem. Soc. Rev.*, 2013, **42**(9), 3689–3707, DOI: 10.1039/C3CS35488A.

35 A. F. Silva, A. Fernandes, P. Neves, M. M. Antunes, S. M. Rocha, M. F. Ribeiro, C. M. Silva and A. A. Valente, Mesostructured Catalysts Based on the BEA Topology for Olefin Oligomerisation, *ChemCatChem*, 2018, **10**(13), 2741–2754, DOI: 10.1002/cctc.201701597.

36 M. Ravi, V. L. Sushkevich and J. A. van Bokhoven, Towards a Better Understanding of Lewis Acidic Aluminium in Zeolites, *Nat. Mater.*, 2020, **19**(10), 1047–1056, DOI: 10.1038/s41563-020-0751-3.

37 Q. Li, K. C. Hunter and A. L. L. East, A Theoretical Comparison of Lewis Acid *vs.* Bronsted Acid Catalysis for *n*-Hexane → Propane + Propene, *J. Phys. Chem. A*, 2005, **109**(28), 6223–6231, DOI: 10.1021/jp050223h.

38 A. Corma and B. W. Wojciechowski, The Chemistry of Catalytic Cracking, *Catal. Rev.: Sci. Eng.*, 1985, **27**(1), 29–150, DOI: 10.1080/01614948509342358.

39 P. del Campo, C. Martínez and A. Corma, Activation and Conversion of Alkanes in the Confined Space of Zeolite-Type Materials, *Chem. Soc. Rev.*, 2021, **50**(15), 8511–8595, DOI: 10.1039/D0CS01459A.

40 S. Schallmoser, T. Ikuno, M. F. Wagenhofer, R. Kolvenbach, G. L. Haller, M. Sanchez-Sanchez and J. A. Lercher, Impact of the Local Environment of Brønsted Acid Sites in ZSM-5 on the Catalytic Activity in *n*-Pentane Cracking, *J. Catal.*, 2014, **316**, 93–102, DOI: 10.1016/j.jcat.2014.05.004.

41 IZA, *Zeolite structures database*, http://www.iza-structure.org/databases/.

42 V. Blay, B. Louis, R. Miravalles, T. Yokoi, K. A. Peccatiello, M. Clough and B. Yilmaz, Engineering Zeolites for Catalytic Cracking to Light Olefins, *ACS Catal.*, 2017, **7**(10), 6542–6566, DOI: 10.1021/acscatal.7b02011.

43 P. Magnoux, P. Cartraud, S. Mignard and M. Guisnet, Coking, Aging, and Regeneration of Zeolites: III. Comparison of the Deactivation Modes of H-Mordenite, HZSM-5, and HY during *n*-Heptane Cracking, *J. Catal.*, 1987, **106**(1), 242–250, DOI: 10.1016/0021-9517(87)90228-4.

44 L. Sørum, M. G. Grønli and J. E. Hustad, Pyrolysis Characteristics and Kinetics of Municipal Solid Wastes, *Fuel*, 2001, **80**(9), 1217–1227, DOI: 10.1016/S0016-2361(00)00218-0.

45 P. Cnudde, K. De Wispelaere, L. Vanduyfhuys, R. Demuynck, J. Van der Mynsbrugge, M. Waroquier and V. Van Speybroeck, How Chain Length and Branching Influence

the Alkene Cracking Reactivity on H-ZSM-5, *ACS Catal.*, 2018, **8**(10), 9579–9595, DOI: 10.1021/acscatal.8b01779.

46 A. A. Brillis and G. Manos, Deactivation Studies During Catalytic Cracking of C8 Aliphatic Hydrocarbons over Ultrastable Y-Zeolite. Conversion and Product Yield Profiles with Time Onstream, *Catal. Lett.*, 2004, **91**(4), 185–191, DOI: 10.1023/B:CATL.0000007153.29038.32.

47 T. J. H. Vlugt, W. Zhu, F. Kapteijn, J. A. Moulijn, B. Smit and R. Krishna, Adsorption of Linear and Branched Alkanes in the Zeolite Silicalite-1, *J. Am. Chem. Soc.*, 1998, **120**(22), 5599–5600, DOI: 10.1021/ja974336t.

48 C. L. Cavalcante Jr and D. M. Ruthven, Adsorption of Branched and Cyclic Paraffins in Silicalite. 2. Kinetics, *Ind. Eng. Chem. Res.*, 1995, **34**(1), 185–191, DOI: 10.1021/ie00040a018.

49 R. Krishna and J. M. van Baten, Diffusion of Hydrocarbon Mixtures in MFI Zeolite: Influence of Intersection Blocking, *Chem. Eng. J.*, 2008, **140**(1), 614–620, DOI: 10.1016/j.cej.2007.11.026.

50 P. Wang, S. Wang, Y. Yue, T. Wang and X. Bao, Effects of Acidity and Topology of Zeolites on the *n*-Alkane Conversion at Low Reaction Temperatures, *Microporous Mesoporous Mater.*, 2020, **292**, 109748, DOI: 10.1016/j.micromeso.2019.109748.

51 A. A. Brillis and G. Manos, Coke Formation during Catalytic Cracking of C8 Aliphatic Hydrocarbons over Ultrastable Y Zeolite, *Ind. Eng. Chem. Res.*, 2003, **42**(11), 2292–2298, DOI: 10.1021/ie020460w.

52 D. Zhao, X. Wang, J. B. Miller and G. W. Huber, The Chemistry and Kinetics of Polyethylene Pyrolysis: A Process to Produce Fuels and Chemicals, *ChemSusChem*, 2020, **13**(7), 1764–1774, DOI: 10.1002/cssc.201903434.

53 E. Kiran and J. K. Gillham, Pyrolysis-Molecular Weight Chromatography: A New on-Line System for Analysis of Polymers. II. Thermal Decomposition of Polyolefins: Polyethylene, Polypropylene, Polyisobutylene, *J. Appl. Polym. Sci.*, 1976, **20**(8), 2045–2068, DOI: 10.1002/app.1976.070200803.

54 M. Guisnet and L. Pinard, Characterization of Acid-Base Catalysts through Model Reactions, *Catal. Rev.: Sci. Eng.*, 2018, **60**(3), 337–436, DOI: 10.1080/01614940.2018.1446683.

55 M. Guisnet, P. Magnoux and D. Martin, Roles of Acidity and Pore Structure in the Deactivation of Zeolites by Carbonaceous Deposits, *Stud. Surf. Sci. Catal.*, 1997, **111**, 1–19, DOI: 10.1016/s0167-2991(97)80138-3.

56 M. L. Sarazen, E. Doskocil and E. Iglesia, Catalysis on Solid Acids: Mechanism and Catalyst Descriptors in Oligomerization Reactions of Light Alkenes, *J. Catal.*, 2016, **344**, 553–569, DOI: 10.1016/j.jcat.2016.10.010.

57 L. F. Lin, S. F. Zhao, D. W. Zhang, H. Fan, Y. M. Liu and M. Y. He, Acid Strength Controlled Reaction Pathways for the Catalytic Cracking of 1-Pentene to Propene over ZSM-5, *ACS Catal.*, 2015, **5**(7), 4048–4059, DOI: 10.1021/cs501967r.

58 L. D. T. Câmara, H. S. Cerqueira, D. A. G. Aranda and K. Rajagopal, Application of a Three-Dimensional Network Model to the Coke Formation in FAU, MFI and BEA Zeolites, *Catal. Today*, 2004, **98**(1), 309–315, DOI: 10.1016/j.cattod.2004.07.044.

59 J. K. Y. Kiang, P. C. Uden and J. C. W. Chien, Polymer Reactions—Part VII: Thermal Pyrolysis of Polypropylene, *Polym. Degrad. Stab.*, 1980, **2**(2), 113–127, DOI: 10.1016/0141-3910(80)90033-6.

60 E. T. C. Vogt and B. M. Weckhuysen, Fluid Catalytic Cracking: Recent Developments on the Grand Old Lady of Zeolite Catalysis, *Chem. Soc. Rev.*, 2015, **44**(20), 7342–7370, DOI: 10.1039/C5CS00376H.

61 A. Akah and M. Al-Ghrami, Maximizing Propylene Production *via* FCC Technology, *Appl. Petrochem. Res.*, 2015, **5**(4), 377–392, DOI: 10.1007/s13203-015-0104-3.

62 B. S. Greensfelder, H. H. Voge and G. M. Good, Catalytic and Thermal Cracking of Pure Hydrocarbons: Mechanisms of Reaction, *Ind. Eng. Chem.*, 1949, **41**(11), 2573–2584, DOI: 10.1021/ie50479a043.

63 N. Batalha, L. Pinard, S. Morisset, J. L. Lemberton, Y. Pouilloux, M. Guisnet, F. Lemos and F. R. Ribeiro, Comparison of the Performances of Pt/HBEA Nano Dispersed over Alumina and Pt/ZSM-22 Catalysts in *n*-Hexadecane Hydroisomerization, *React. Kinet., Mech. Catal.*, 2012, **107**(2), 285–294, DOI: 10.1007/s11144-012-0488-3.

64 M. Guisnet, "Ideal" Bifunctional Catalysis over Pt-Acid Zeolites, *Catal. Today*, 2013, **218–219**, 123–134, DOI: 10.1016/j.cattod.2013.04.028.

65 G. Giannetto, F. Alvarez, F. R. Ribeiro, G. Pérot and M. Guisnet, Hydroisomerization and Hydrocracking of Alkanes. 6. Influence of the Pore Structure on the Selectivity of Pt-Zeolite, in *Guidelines for Mastering the Properties of Molecular Sieves: Relationship between the Physicochemical Properties of Zeolitic Systems and Their Low Dimensionality*, ed. D. Barthomeuf, E. G. Derouane and W. Hölderich, Springer US, Boston, MA, 1990, pp. 355–363, DOI: 10.1007/978-1-4684-5787-2_21.

66 C. Liu, Y. Deng, Y. Pan, Y. Gu, B. Qiao and X. Gao, Effect of ZSM-5 on the Aromatization Performance in Cracking Catalyst, *J. Mol. Catal. A: Chem.*, 2004, **215**(1), 195–199, DOI: 10.1016/j.molcata.2004.02.001.

67 N. Viswanadham, J. K. Gupta, G. M. Dhar and M. O. Garg, Effect of Synthesis Methods and Modification Treatments of ZSM-5 on Light Alkane Aromatization, *Energy Fuels*, 2006, **20**(5), 1806–1814, DOI: 10.1021/ef0601770.

68 A. Beuque, M. Barreau, E. Berrier, J.-F. Paul, N. Batalha, A. Sachse and L. Pinard, Transformation of Dilute Ethylene at High Temperature on Micro- and Nano-Sized H-ZSM-5 Zeolites, *Catalysts*, 2021, **11**(2), 282, DOI: 10.3390/catal11020282.

69 H. J. Park, H. S. Heo, J.-K. Jeon, J. Kim, R. Ryoo, K.-E. Jeong and Y.-K. Park, Highly Valuable Chemicals Production from Catalytic Upgrading of Radiata Pine Sawdust-Derived Pyrolytic Vapors over Mesoporous MFI Zeolites, *Appl. Catal., B*, 2010, **95**(3), 365–373, DOI: 10.1016/j.apcatb.2010.01.015.

70 B. Luna-Murillo, M. Pala, A. L. Paioni, M. Baldus, F. Ronsse, W. Prins, P. C. A. Bruijnincx and B. M. Weckhuysen,

Catalytic Fast Pyrolysis of Biomass: Catalyst Characterization Reveals the Feed-Dependent Deactivation of a Technical ZSM-5-Based Catalyst, *ACS Sustainable Chem. Eng.*, 2021, **9**(1), 291–304, DOI: 10.1021/acssuschemeng.0c07153.

71 X. Chen and Y. Yan, Study on the Technology of Thermal Cracking of Paraffin to Alpha Olefins, *J. Anal. Appl. Pyrolysis*, 2008, **81**(1), 106–112, DOI: 10.1016/j.jaap.2007.09.009.

72 M. Guisnet, L. Costa and F. R. Ribeiro, Prevention of Zeolite Deactivation by Coking, *J. Mol. Catal. A: Chem.*, 2009, **305**(1), 69–83, DOI: 10.1016/j.molcata.2008.11.012.

73 M. Guisnet and P. Magnoux, Coking and Deactivation of Zeolites: Influence of the Pore Structure, *Appl. Catal.*, 1989, **54**(1), 1–27, DOI: 10.1016/S0166-9834(00)82350-7.

74 X. Hou, L. Zhao and Z. Diao, Roles of Alkenes and Coke Formation in the Deactivation of ZSM-5 Zeolites During *n*-Pentane Catalytic Cracking, *Catal. Lett.*, 2020, **150**(9), 2716–2725, DOI: 10.1007/s10562-020-03173-4.

75 J. Jae, G. A. Tompsett, A. J. Foster, K. D. Hammond, S. M. Auerbach, R. F. Lobo and G. W. Huber, Investigation into the Shape Selectivity of Zeolite Catalysts for Biomass Conversion, *J. Catal.*, 2011, **279**(2), 257–268, DOI: 10.1016/j.jcat.2011.01.019.

76 P. C. Mihindou-Koumba, H. S. Cerqueira, P. Magnoux and M. Guisnet, Methylcyclohexane Transformation over HFAU, HBEA, and HMFI Zeolites: II. Deactivation and Coke Formation, *Ind. Eng. Chem. Res.*, 2001, **40**(4), 1042–1051, DOI: 10.1021/ie0007318.

77 B. Xu, C. Sievers, S. B. Hong, R. Prins and J. A. van Bokhoven, Catalytic Activity of Brønsted Acid Sites in Zeolites: Intrinsic Activity, Rate-Limiting Step, and Influence of the Local Structure of the Acid Sites, *J. Catal.*, 2006, **244**(2), 163–168, DOI: 10.1016/j.jcat.2006.08.022.

78 M. M. Pereira, F. M. Santos, A. V. Silva, N. Batalha, F. Junior Ferreira da Silva Henrique, P. M. Esteves and B. Louis, Insights into the Role of Framework and Nonframework Aluminum in the Protolytic Reaction of Carbon–Carbon and Tertiary Carbon–Hydrogen Bonds of Isobutane, *J. Phys. Chem. C*, 2021, **125**(21), 11636–11647, DOI: 10.1021/acs.jpcc.1c01989.

